CHAPTER 2

Demography

With Total Fertility Rate (TFR) at 1.8, the population of the State has reached sub replacement level. The registration of births and deaths under the Codigo do Registo Civil do Estado da India precedes the registration of Births and Deaths Act, 1969. The institutional mechanism for registration is well established and the vital rates are available based on 100% registration in contrast to the schemes of the sample registration system of the RGI which provide data on the subject. Statements showing Registration of Births & Deaths from 1995 to 2005 (quinquennial) and for the years 2006 & 2007 in Goa by district, taluka, sex and period of registration is given in Annexure 10 & 11 respectively.

The Birth Rate, Death Rate and Infant Mortality Rate of Goa, from 1991 to 2006 is given in table 2.1.

Table 2.1 : Birth Rate, Death Rate and Infant Mortality Rate - Goa

Year	Birth Rate	Death Rate	Infant Mortality Rate
1991	19.50	7.46	18.76
1992	18.59	7.44	18.06
1993	17.88	7.54	18.43
1994	18.00	7.47	15.66
1995	17.82	7.07	14.39
1996	18.00	7.40	17.25
1997	18.34	7.37	16.92
1998	17.63	8.26	17.21
1999	17.40	7.47	16.42
2000	17.55	7.49	12.46
2001 (P)	15.95	7.46	11.91
2002 (P)	16.59	7.81	12.48
2003 (P)	15.84	7.68	13.17
2004 (P)	15.86	7.70	14.70
2005 (P)	15.84	7.77	14.12
2006 (P)	15.10	7.40	15.00

P : Provisional.

Source : Returns collected by the O/o The Chief Registrar of Births and Deaths.

Note : Above vital rates are based on 100% Registration of Births & Deaths in the State.

The natural growth of population i.e Birth Rate less Death Rate has progressively declined from 1.2% in 1991 to 0.8% in 2001. The average natural growth during the decade is 1.06% per annum. Also, the IMR has declined from 18.76 in 1991 to 13-15 in recent years. The present level achieved by the State is the best among all the States and Union Territories in the country. (Annexure 12). Monthwise registration of infant deaths in Goa from 2000 to 2005 (quinquennial) and for the years 2006 & 2007 by sex is given in Annexure 13. It is seen that there has been continuous decline in infant deaths.

a. Population Censuses : Census 2001

In Goa the erstwhile Portuguese Administration was reported to be conducting Population Census and the history dates back to 1850. The last census conducted by Portuguese Administration in Goa was in 1960. From the year 1971 onwards, the population count in Goa has been aligned as per the pattern of decadal census prevalent in the rest of the country. So far, four population censuses have been conducted in the years 1971, 1981, 1991 and 2001. The last population census was held in March 2001. As per the final results released by the Registrar General of India, the population of Goa is 1347668 persons. The population of Goa and the decadal growth rate during 1900 to 2001 Census is given in table 2.2.

Table 2.2 : Population of Goa and Decadal Growth Rate 1900-2001

Sl. No.	Year	North Goa District	South Goa District	Goa	Decadal growth (%)
1	1900	294074	181439	475513	
2	1910	306323	180429	486752	+2.36
3	1921	288039	181455	469494	-3.55
4	1931	313614	191667	505281	+7.62
5	1940	336628	204297	540925	+7.05
6	1950	330874	216574	547448	+1.21
7	1960	349667	240330	589997	+7.77
8	1971	458312	336808	795120	+34.77
9	1981	568021	439728	1007749	+26.74
10	1991	664804	504989	1169793	+16.08
11	2001	758573	589095	1347668	+15.21

Out of the total population of 13,47,668, North Goa District accounts for 43.71% and South Goa District for 56.29%. The decadal growth rate of population has declined from 16.08 during 1991 to 15.21% during 2001.

b. Age Composition

Age is one of the most important demographic characteristics for effective policy formulations and planning. The age composition by residence, five year age group and sex, for the State of Goa is given in tables 2.3

Table 2.3 : Population in Five Year Age Group by Sex for the State

Sl. No.	Age group	Persons	Total Males	Females
1	**All ages**	**1347668**	**687248**	**660420**
2	0-4	103823	53624	50199
3	5-9	107390	54996	52394
4	10-14	120013	61304	58709
5	15-19	125031	64970	60061
6	20-24	142016	75652	66364
7	25-29	138614	70745	67869
8	30-34	117544	61304	56240
9	35-39	104565	53245	51320
10	40-44	82539	44065	38474
11	45-49	75632	38066	37566
12	50-54	64311	32882	31429
13	55-59	46905	23063	23842
14	60-64	42394	19972	22422
15	65-69	28168	12501	15667
16	70-74	20484	8650	11834
17	75-79	10315	4279	6036
18	80+	10912	4274	6638
19	Age not stated	7012	3656	3356

The male-female differential in the age distribution of population indicates that male population exceeds the female population in all the age groups from 0-4 years to 50-54 years. In the age group 0-4 years, male proportion is higher by 1.89% whereas in combined age group 0-14 years, it is higher by 2.60%. Yet higher proportion of 6.78% and 6.54% is in respect of age groups 40-44 years and 20-24 years respectively.

c. Rural Urban composition

Post liberation period witnessed heavy urbanization in Goa. As per 2001 census, 49.75% population lives in urban areas. The rural urban composition of population in five year age groups by residence and sex is given in table 2.4.

Table 2.4 : Population in Five Year Age Group by Residence and Sex for the State

Goa State						
Age group	Rural			Urban		
	Persons	Males	Females	Persons	Males	Females
All ages	**677091**	**340545**	**336546**	**670577**	**346703**	**323874**
0-4	51505	26426	25079	52318	27198	25120
5-9	51047	26108	24939	56343	28888	27455
10-14	59453	30234	29219	60560	31070	29490
15-19	64091	32570	31521	60940	32400	28540
20-24	72435	37803	34632	69581	37849	31732
25-29	69748	35778	33970	68866	34967	33899
30-34	57642	30168	27474	59902	31136	28766
35-39	50703	25637	25066	53862	27608	26254
40-44	40079	21026	19053	42460	23039	19421
45-49	37570	18549	19021	38062	19517	18545
50-54	31947	15678	16269	32364	17204	15160
55-59	24471	11370	13101	22434	11693	10741
60-64	23093	10591	12502	19301	9381	9920
65-69	15502	6718	8784	12666	5783	6883
70-74	11429	4761	6668	9055	3889	5166
75-79	5583	2275	3308	4732	2004	2728
80+	5866	2287	3579	5046	1987	3059
Age not stated	4927	2566	2361	2085	1090	995

The proportion of female population in age groups from 60-64 years to 80+ years is much higher than the male population except in the age group 60-64 years where it is 1.66% more. In the age groups 65-69 years to 80+ years, this proportion varies from 4.28% to 21.66%. A higher proportion of females 21.66% is in the age group 75-79 years followed by 17.04% in the age group 70-74 years.

The proportion of children in the age group 0-4 years and also in the combined age group 0-14 years is higher in urban areas than in rural areas for both males and females. A higher proportion of males in the working age group 15-59 years live in urban areas as compared to rural areas. Proportion of aged persons (60 years and above) is higher in rural areas than the corresponding share in urban areas.

d. Sex ratio

The sex ratio of population in different talukas of Goa and of child population in the age group of 0-6 years is given in table 2.5. The sex ratio in terms of females per thousand males for the period from 1900 to 2001 is given in table 2.6.

Table 2.5 : Female Sex Ratio in different Talukas of Goa
Population Census 2001

State/ District/ Taluka	Female sex ratio					
	Total population			Age-group 0-6 years		
	Rural	Urban	Total	Rural	Urban	Total
Goa	**988**	**934**	**961**	**952**	**924**	**938**
North Goa	**970**	**931**	**953**	**939**	**935**	**938**
Pernem	943	942	943	921	827	907
Bardez	996	929	956	926	924	925
Tiswadi	1013	947	969	932	972	958
Bicholim	946	938	943	939	914	928
Satari	961	964	962	935	980	941
Ponda	959	893	937	970	921	953
South Goa	**1018**	**937**	**972**	**972**	**913**	**937**
Mormugao	1026	868	893	917	889	893
Salcete	1065	986	1019	1000	928	957
Quepem	975	970	973	937	909	924
Sanguem	964	980	967	997	911	981
Canacona	996	921	975	925	998	944

Table 2.6 : Female Sex Ratio of Goa 1900-2001

Sl. No.	Year	Female sex ratio
1	1900	1091
2	1910	1108
3	1921	1120
4	1931	1088
5	1940	1084
6	1950	1128
7	1960	1066
8	1971	981
9	1981	975
10	1991	967
11	2001	961

e. Literacy

The population in age group 0-6 years is excluded for the purpose of calculation of literacy rate as per census definition of literacy. Out of the total population of 1347668, the population in the age group 0-6 years is 145968. Literacy rates by sex during 1991 and 2001 census is given in table 2.7.

Table 2.7 : Literacy Rate of Goa by Sex (per cent)

	1991 census	2001 census
Total	**75.51**	**82.01**
Male	83.64	88.42
Female	67.09	75.37

Literacy rate both for male and female has increased in the year 2001 as compared to 1991. Overall literacy in urban areas is higher than in rural areas. However, in Pernem and Ponda while male literacy in rural areas is more than in urban areas, in Bardez the overall literacy in rural areas as also the male literacy and female literacy in rural areas is higher than the corresponding literacy in urban areas. Literacy rate in different talukas of Goa by sex is given in table 2.8

Table 2.8 : Literacy Rates in different Talukas of Goa Population Census 2001 (Per cent)

State/District/ Taluka	Rural			Urban			Total		
	P	M	F	P	M	F	P	M	F
Goa	**79.7**	**87.4**	**71.9**	**84.4**	**89.5**	**79.0**	**82.0**	**88.4**	**75.4**
North Goa	**82.0**	**89.9**	**73.8**	**85.4**	**90.2**	**80.3**	**83.5**	**90.0**	**76.7**
Pernem	80.6	89.5	71.1	82.1	89.0	74.8	80.8	89.5	71.6
Bardez	86.6	92.1	81.2	84.8	89.1	80.2	85.5	90.3	80.6
Tiswadi	82.4	89.4	75.6	86.3	91.1	81.1	84.9	90.5	79.2
Bicholim	81.7	90.6	72.3	86.8	92.9	80.3	83.8	91.5	75.6
Satari	74.5	84.7	63.9	88.3	94.2	82.1	76.4	86.0	66.3
Ponda	82.0	90.6	73.0	84.7	89.2	79.7	82.9	90.1	75.1
South Goa	**76.0**	**83.2**	**69.0**	**83.3**	**88.7**	**77.6**	**80.1**	**86.3**	**73.7**
Mormugao	79.0	86.0	72.4	83.7	89.6	77.0	82.9	89.0	76.1
Salcete	79.5	85.5	74.0	83.7	88.3	79.1	81.9	87.1	76.8
Quepem	69.2	76.9	61.4	81.4	87.6	75.2	74.8	81.8	67.7
Sanguem	74.2	83.5	64.5	82.6	89.9	75.3	75.7	84.6	66.4
Canacona	72.8	80.6	64.9	80.1	85.9	73.7	74.8	82.1	67.2

f. Religion - wise composition of population

The religion wise composition of population of Goa during 1960,1971,1981,1991 and 2001 is given in table 2.9. The population classified by religion has been showing an increasing trend. During the last four decades, the Hindu population has increased by two and half times, Christian population by one and half times, Muslim population by about eight times and others by over twenty six times. Given that the increase in population is not on account of increase in fertility, it may be attributed to inmigration.

Table 2.9 : Religion-wise Composition of population from 1960 to 2001 Goa

Sl. No.	Religion based classification	1960	1971	1981	1991	2001
1	Hindu	353526	496389	646986	756621	886551
2	Christian	224612	270126	315902	349225	359568
3	Muslim	11505	26480	41317	61455	92210
4	Others including RNS*	354	2125	3544	2492	9339
5	**Total**	**589997**	**795120**	**1007749**	**1169793**	**1347668**

* RNS: Religion Not Stated

g. Migration

Migration refers to the movement of people from one place to another, especially in order to find work or to live there for a short time. It is a major factor in changing the size and structure of the population and the process impacts the area to which migrants move in as also the area from which they move out. The Total Fertility Rate (TFR) in the State being 1.8 (2005-06), which is below the replacement rate, i.e. 2.1, it would follow that the increase in population is not on account of reproduction. However, whether the given TFR will eventually translate into lower population will depend on the age structure of the population and other variables and it is an issue that would require expert analysis.

The Census of India collects migration data and publishes detailed migration tables, albeit with significant time lag. The Census provides migration data on all migration streams by concepts of place of birth and place of last residence. This data alongwith reasons of migration and duration of stay in the current place of residence provides useful insight for studying migration dynamics of population. These tables are based mainly on the response to questions regarding place of birth, place of last residence, duration of residence at the place of enumeration and reason for migration. The migratory movement of a person by place of last residence showing intra-district, inter-district, intra-state, inter-state and also migration from abroad is captured at the place of enumeration at the State level.

The Census of India 2001 has listed 7,85,020 migrants classified by the place of last residence at the place of enumeration. Of these the last residence of 5,46,392 migrants is within the State but outside the place of enumeration, thus the last residence of the balance migrants is in States outside Goa. The State-wise break-up of the number of these migrants classified by place of last residence is as follows:

Place of last Residence	Number of Migrants
Andhra Pradesh	6355
Arunachal Pradesh	23
Assam	449
Bihar	4066
Gujarat	3041
Haryana	699
Himachal Pradesh	324
Jammu & Kashmir	881
Karnataka	112116
Kerala	10251
Madhya Pradesh	1579
Maharashtra	58750
Manipur	68
Meghalaya	44
Mizoram	9
Nagaland	34
Orissa	2148
Punjab	678
Rajasthan	2550
Sikkim	15
Tamilnadu	5041
Tripura	25
Uttar Pradesh	11532
West Bengal	3272
Andaman & Nicobar	325
Chandigarh	174
Dadra & Nagar Haveli	41
Daman & Diu	318
Delhi	1687
Lakshadweep	0
Pondicherry	114
Chattisgarh	394
Jharkhand	1444
Uttarakhand	422
Total	**228869**
Outside India	*9759*
GRAND TOTAL	**238628**

The Census takes into account only the last migratory movement of a person even though he/she might have migrated several times in life. However, assuming that all the migrants, whose last residence is in States outside Goa, were all settled in Goa at the point of enumeration, these would constitute 17% of the total population, as per the 2001 Census. The decadal growth of population during 1991-2001 being 15.21%, of which the average natural increase is 10.6%, it would follow that the balance 4.6% may be attributed to in-migration. Globalization is contributing to migration on a significant scale and Goa may not be an exception to it. In the context of the formulation of the

Eleventh Five Year Plan, the Planning Commission has estimated the elasticity of employment with respect to the GSDP for each of the major States and for the country as a whole. Accordingly, the additional employment opportunities likely to be generated have been calculated and the model hypothesizes that considerable cross-migration will be involved if the State-wise employment needs and opportunities are to be met.

h. Workers & Non workers

As per 2001 Census, 'work' is defined as participation in any economically productive activity with or without compensation, wages or profit. The two categories of workers 'Main Workers and Marginal Workers' together are called 'Total Workers'. A person who has not worked at all is classified as a 'Non-Worker'. Distribution of main workers, marginal workers and non-workers as per 2001 Census is given in table 2.10.

Table 2.10 : Sex-wise distribution of Workers & Non-Workers : Census 2001

Sl. No.	Classification	Persons	Males	Females
1.	Main Workers	425305	326993	98312
2.	Marginal Workers	97550	48225	49325
3.	Total Workers (Main + Marginal)	522855	375218	147637
4.	Non-workers	824813	312030	512783

Main Workers and Marginal Workers are further classified under four broad categories viz. Cultivators, Agricultural labourers, Household Industry Workers and Other Workers. Of the total workers, 50395 are cultivators, 35806 are agricultural labourers, 14746 are household industry workers and 421908 are other workers. In terms of percentage, cultivators form 9.64%, agricultural labourers 6.85%, household industry workers 2.82% and other category of workers 80.69%.

The Distribution of Main & Marginal Workers under various categories of workers by sex is given in table 2.11.

Table 2.11 : Main & Marginal Workers by Sex 2001

Category	Sex	Total Workers		Main Workers		Marginal Workers	
		Rural	Urban	Rural	Urban	Rural	Urban
1. Cultivators	Male	23550	2158	16875	1490	6675	668
	Female	22335	2352	10187	1126	12148	1226
2. Agricultural Labourers	Male	13592	2378	5854	1004	7738	1374
	Female	17484	2352	4359	625	13125	1727
3. Household Industry	Male	5093	3898	4021	3222	1072	676
	Female	3633	2122	1683	1313	1950	809
4. Other Workers	Male	143413	181136	127319	167208	16094	13928
	Female	45352	52007	34617	44402	10735	7605
5. Total Workers	Male	185648	189570	154069	172924	31579	16646
	Female	88804	58833	50846	47466	37958	11367

The total workers (main and marginal) together form 38.80 per cent of the total population. The proportion of male workers to total workers is 71.76 per cent. Workers in the rural areas comprise 52.49 per cent of the total workers.

The Work Participation Rate (WPR) during 1991 and 2001 Censuses are given in table 2.12.

Table 2.12 : Work Participation Rate : 1991 and 2001

Sl. No.	Persons/ Males Females	Main workers	Marginal workers	Total workers
	Persons			
	1991	32.8	2.5	35.3
1	2001	31.6	7.3	38.8
	Males			
	1991	48.3	1.3	49.6
2	2001	47.6	7.0	54.6
	Females			
	1991	16.8	3.8	20.5
3	2001	14.9	7.5	22.4